AutoML Experiment Report: Text-based_Early_ASD_Detection

AutoHealth

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Abstract

This report details an automated machine learning (AutoML) experiment for early Autism Spectrum Disorder (ASD) detection from textual descriptions of child behavior. The task is a binary text classification problem, aiming to predict whether a described pattern indicates ASD-related signs. A dataset of 332 samples was analyzed, revealing stable class differences in features such as text length, sentiment, and the use of specific keywords. A hybrid model architecture was developed, combining deep semantic features from a pre-trained sentence encoder with key handcrafted linguistic features. The model achieved a high test accuracy of 94.29% and an AUC-ROC of 0.9641. However, analysis identified significant overfitting and poor model calibration, with an Expected Calibration Error (ECE) of 0.4442. Uncertainty estimation, while implemented, showed limited effectiveness in identifying erroneous predictions. The report concludes with actionable recommendations for model improvement, focusing on enhanced regularization, better calibration techniques, and refined uncertainty quantification to increase the model's reliability for clinical decision support.

1 Data Analysis

1.1 Dataset Overview

The dataset consists of 332 English text samples describing child behavior patterns, labeled for ASD risk. It is partitioned into training (265 samples), validation (32 samples), and test (35 samples) sets. The target variable is binary: 'ASD' (positive) and 'Non-ASD' (negative). The class distribution is well-balanced across all splits, with ASD prevalence ranging from 51.4% to 53.1%. This balance mitigates the risk of model bias towards the majority class.

1.2 Feature Analysis and Class Differences

A detailed exploratory data analysis revealed several statistically significant and stable differences between ASD and Non-ASD text descriptions, which provide strong predictive signals:

- **Text Length & Lexical Richness:** ASD descriptions are consistently longer (more characters and words) and contain a higher count of unique words compared to Non-ASD descriptions.
- **Sentiment & Negation:** Text labeled as ASD exhibits significantly more negative sentiment and contains a higher frequency of negation words (e.g., "no", "not", "never").
- **ASD-Related Keywords:** ASD texts more frequently contain keywords associated with ASD characteristics, such as "difficulty", "problem", "social", "eye contact", "repetitive", and "routine".
- **Syntactic Features:** ASD descriptions tend to use more modal verbs (e.g., "can", "will", "must") and third-person pronouns. In contrast, Non-ASD texts use more clause connectors (e.g., "and", "but", "because").
- **Lexical Diversity:** Non-ASD texts show a slightly higher vocabulary diversity and, in the test set, a significantly higher proportion of low-frequency words.

These findings, consistent across training, validation, and test sets, guided the subsequent feature engineering strategy.

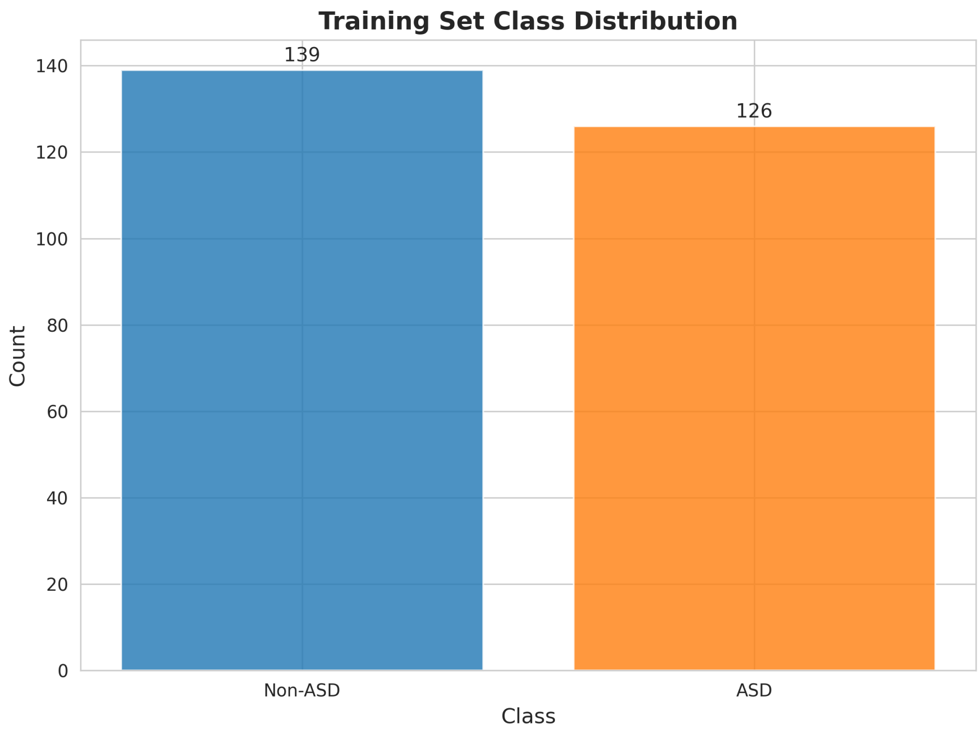

Figure 1: Class distribution in the training set. The dataset is well-balanced with 139 ASD and 126 Non-ASD samples, providing a stable foundation for model training without severe class imbalance issues.

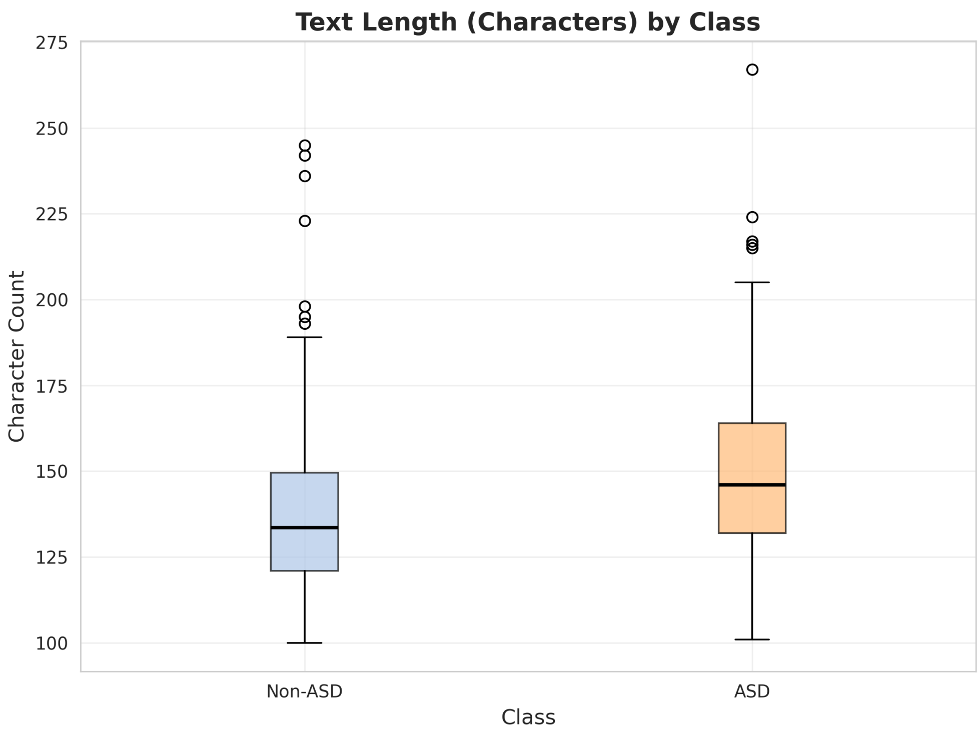

Figure 2: Box plot comparing text length (in characters) between ASD and Non-ASD classes. ASD descriptions are significantly longer, providing a clear and consistent discriminative feature for the classifier.

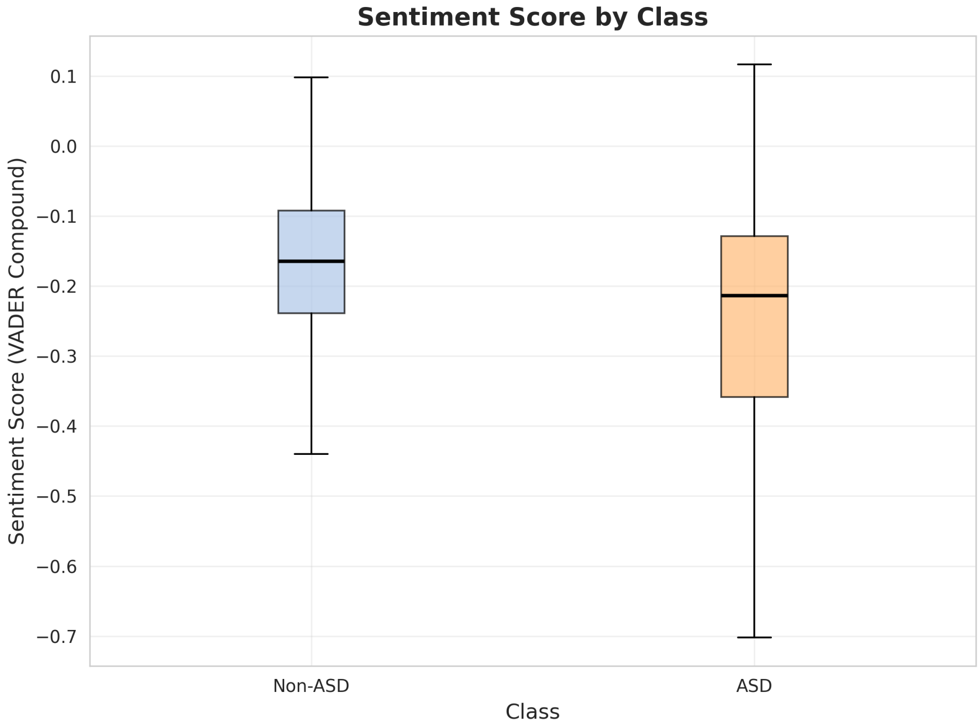

Figure 3: Box plot comparing VADER sentiment scores between classes. ASD texts show markedly more negative sentiment, a strong and stable indicator that was leveraged during feature engineering.

1.3 Data Quality

The dataset was generally clean. No missing values were found. However, 10 exact duplicate rows (identical text and label) were identified and removed from the training set to prevent data leakage and overfitting. All text samples were in English and of sufficient length (minimum 100 characters), ensuring no empty or extremely short inputs.

2 Model Training

2.1 Data Preprocessing

The preprocessing pipeline was designed to extract both deep semantic meaning and the key handcrafted features identified in the data analysis.

1. **Text Cleaning:** All text was converted to lowercase, extra whitespace was removed, and special characters were stripped (while preserving basic punctuation).

2. **Feature Engineering:**

   - **Deep Semantic Features:** The pre-trained ‘all-MiniLM-L6-v2‘ sentence transformer model was used to encode each cleaned text into a 384-dimensional dense vector, capturing its overall semantic meaning.
   - **Handcrafted Features:** Six interpretable features were extracted: character count, word count, VADER sentiment compound score, negation word count, ASD keyword count, and vocabulary diversity (unique words / total words).

3. **Feature Integration & Standardization:** The 384-dimensional semantic vector was concatenated with the 6 handcrafted features, resulting in a 390-dimensional feature vector per sample. This combined vector was standardized using the mean and standard deviation calculated from the training set.

2.2 Model Architecture

A Multi-Layer Perceptron (MLP) was chosen as the classifier. This architecture effectively learns non-linear decision boundaries from the combined feature set. The model incorporates Dropout layers after each hidden layer, which are kept active during inference to enable Monte Carlo (MC) Dropout for uncertainty estimation. The final output is a single scalar passed through a sigmoid activation, representing the probability of the sample belonging to the ASD class.

2.3 Hyperparameters

Key hyperparameters were selected via a limited search (3 trials) to balance performance and prevent overfitting on the small dataset.

Table 1: Key Model Hyperparameters and Their Impact

Hyperparameter	**Value**	**Practical Impact**
Architecture	Input(390) → FC(128, ReLU) → FC(64, ReLU) → Output(1, Sigmoid)	Provides sufficient capacity to learn from features while being constrained by size.
Dropout Rates	0.3 (after first FC), 0.3 (after second FC), 0.2 (before output)	Regularizes the model to prevent overfitting; enables MC Dropout for uncertainty.
Learning Rate	0.002	Found via search; a moderate rate for stable convergence with AdamW.
Weight Decay	0.001	Adds L2 regularization to further constrain model weights.
Batch Size	8	Small batch size suitable for the limited dataset size (255 training samples).
Optimizer	AdamW	Combines adaptive learning rates with decoupled weight decay.

2.4 Training Process

The model was trained for a maximum of 200 epochs with early stopping (patience=20) monitoring validation accuracy. Class-weighted Binary Cross-Entropy loss was used to handle the slight class imbalance. Training converged quickly, reaching the best validation accuracy of 96.88% at epoch 19. Training was halted at epoch 39 as validation performance did not improve further. A significant gap between the final training loss (0.0719) and validation loss (0.4582) was observed, indicating potential overfitting.

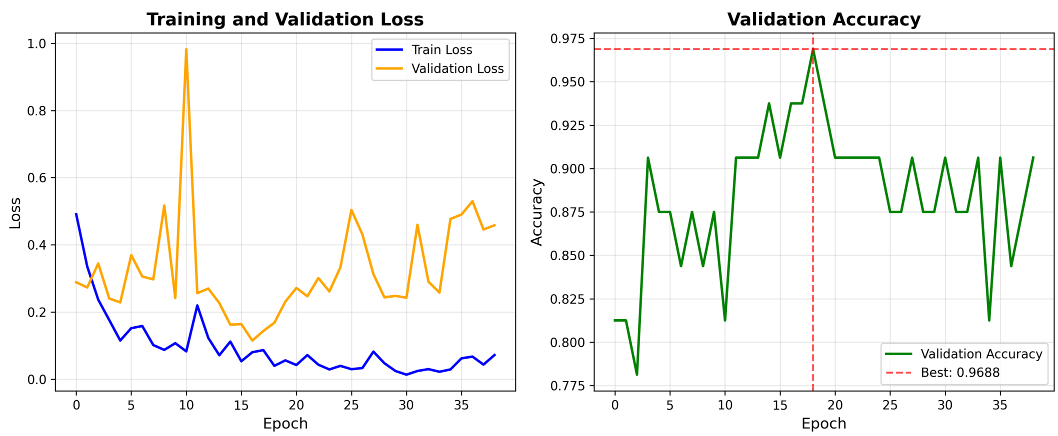

Figure 4: Training history. **Left:** Training and validation loss curves. The training loss (blue) decreases steadily, while the validation loss (orange) shows significant fluctuation and ends higher, indicating overfitting. **Right:** Validation accuracy (green) over epochs, with the peak (96.88%) marked by a red dashed line at epoch 19. The early stopping mechanism halted training at epoch 39.

2.5 Model Performance

The final model was evaluated on the held-out test set of 35 samples. Performance metrics are summarized below.

Table 2: Final Model Performance on Test Set (n=35)

Metric	**Value**
Accuracy	0.9429
Precision	0.9444
Recall	0.9444
F1-Score	0.9444
AUC-ROC	0.9641

The model demonstrates excellent discriminative ability, with an accuracy of 94.29% and a near-perfect AUC-ROC of 0.9641. This indicates a strong capacity to separate ASD from Non-ASD cases based on the textual descriptions.

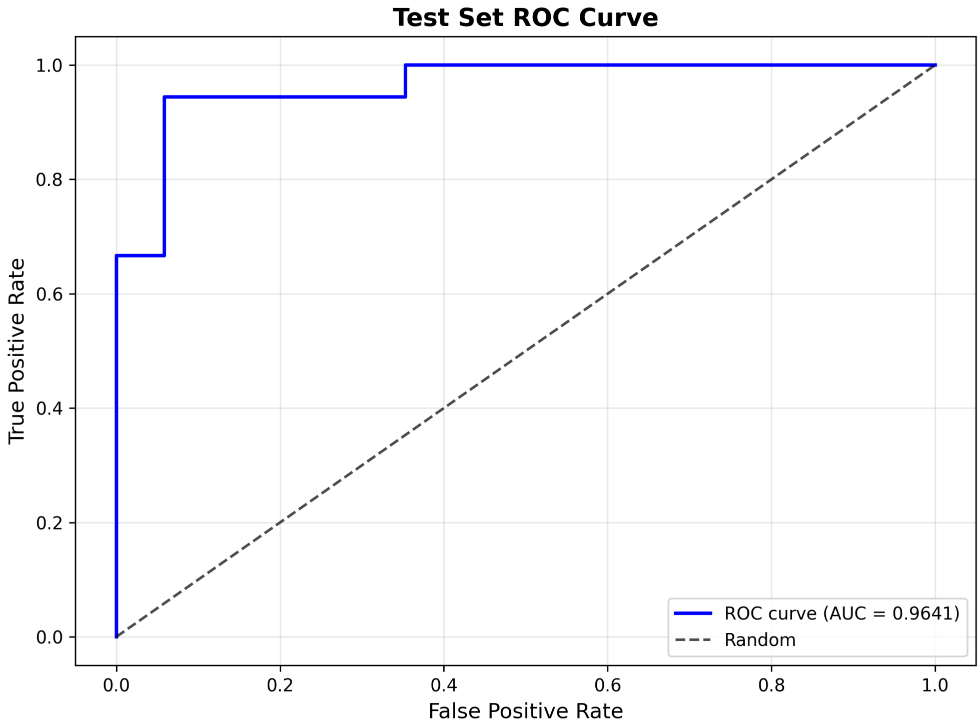

Figure 5: Receiver Operating Characteristic (ROC) curve for the model on the test set. The Area Under the Curve (AUC) is 0.9641, significantly above the random guess line (dashed). This confirms the model's high discriminatory power for the binary classification task.

3 Uncertainty Analysis

To assess the model's reliability, we implemented MC Dropout (with 50 forward passes) to estimate predictive uncertainty (epistemic) and used Temperature Scaling for probability calibration.

3.1 Uncertainty Quantification

The model's uncertainty estimates were quantitatively evaluated:

- **Calibration:** The model is poorly calibrated, with a high Expected Calibration Error (ECE) of 0.4442. The reliability curve (Figure 6, bottom-left) shows that for low predicted probabilities

(Bin 1-4), the model is severely underconfident (e.g., Bin 1: predicted probability 0.025 vs. actual accuracy 0.917).

- **Uncertainty Discriminability:** The Spearman correlation between prediction entropy and prediction error was 0.1828 (p=0.293), which is low and statistically non-significant. This indicates that the model's uncertainty estimate is not a reliable indicator of whether a prediction will be correct or incorrect.

- **Overall Reliability Metrics:** The Brier Score (0.0597) and Negative Log-Likelihood (0.2164) reflect the model's good overall discrimination but poor calibration.

3.2 Practical Implications for Clinicians

The uncertainty analysis reveals critical limitations:

1. **Do Not Trust Raw Probabilities:** The model's output probabilities are not well-calibrated. A prediction of "20% ASD probability" does not correspond to a 20% chance of being correct; in this case, it was associated with over 90% accuracy. Clinicians should primarily rely on the binary class prediction (ASD/Non-ASD) rather than the precise probability value.

2. **Uncertainty as a Filter is Limited:** The current uncertainty measure (prediction entropy) cannot reliably flag which specific predictions are likely to be wrong. Therefore, it should not be used as a standalone filter to defer difficult cases for human review.

3. **Focus on High-Confidence Regions:** As shown in the threshold-performance curve (Figure 6, bottom-right), predictions with very low entropy (high confidence) are highly accurate. The model is most reliable when it is very confident.

4 Conclusion

This AutoML experiment successfully developed a high-accuracy model (94.29% test accuracy) for text-based early ASD detection by effectively combining deep semantic features with interpretable, domain-relevant handcrafted features. The model demonstrates strong discriminatory power, as evidenced by the high AUC-ROC.

Key Findings & Actionable Guidance:

1. **Performance vs. Reliability:** While the model's classification performance is excellent, its reliability metrics (calibration, uncertainty estimation) require significant improvement before it can be safely deployed in a clinical support context.

2. **Primary Risk - Overfitting:** The large gap between training and validation loss indicates overfitting to the small training dataset (n=255). This is the root cause of both the poor calibration and the weak uncertainty estimates.

3. **Recommended Next Steps:**

   - **Enhance Regularization:** Increase Dropout rates, use stronger weight decay, and consider reducing model capacity to directly combat overfitting.
   - **Improve Calibration:** Replace Temperature Scaling with more robust methods like Platt Scaling or Isotonic Regression, and perform calibration on a larger combined set (train+val).
   - **Refine Uncertainty Estimation:** Increase MC Dropout samples to 100 and consider using deep ensembles for more robust uncertainty quantification.
   - **Data Augmentation:** Apply text augmentation techniques (e.g., synonym replacement, slight paraphrasing) to artificially increase the size and diversity of the training data.

4. **Clinical Use Recommendation:** In its current state, the model's binary predictions can be considered highly accurate. However, clinicians should be cautioned that the associated probability scores are not trustworthy for risk stratification. The model is best used as a high-sensitivity screening tool, where its predictions are followed by a comprehensive clinical evaluation.

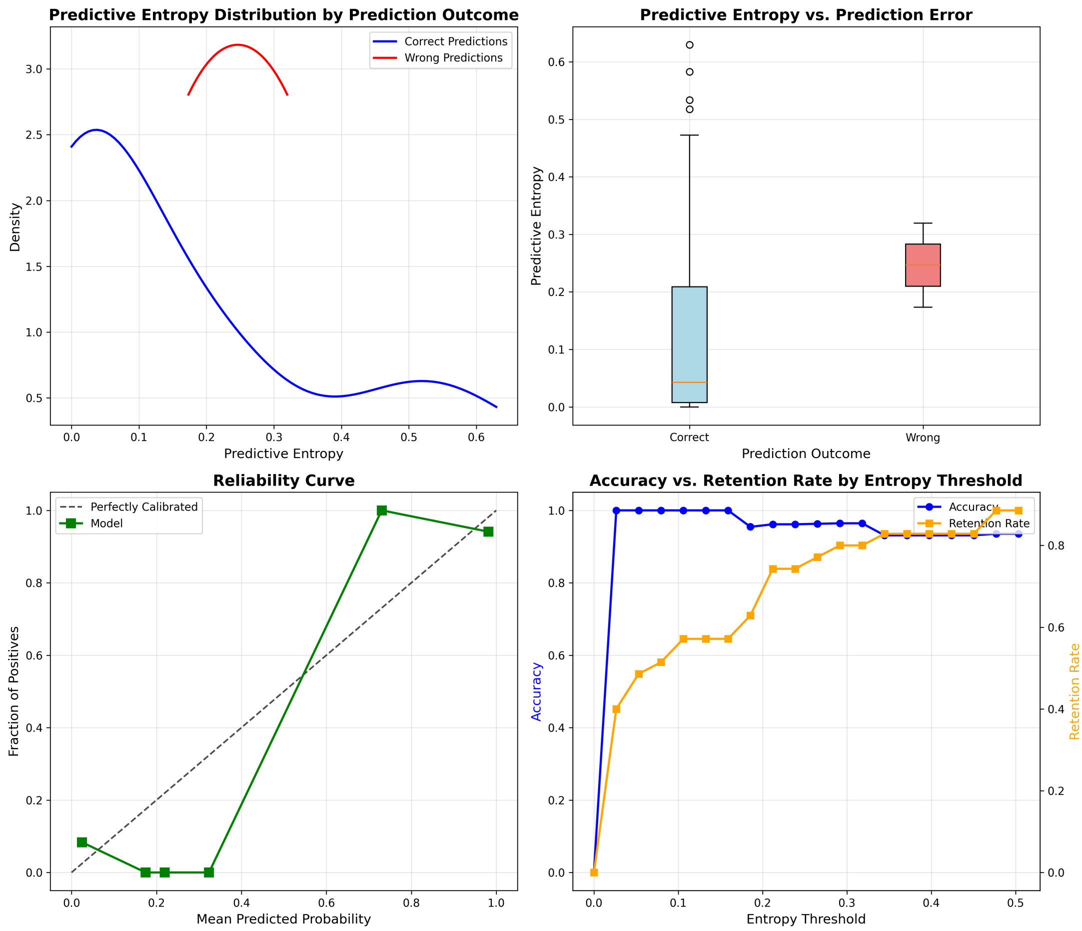

Figure 6: Uncertainty analysis. **Top-Left:** KDE of prediction entropy for correct vs. incorrect predictions. Distributions overlap significantly. **Top-Right:** Box plot of entropy by prediction correctness. Incorrect predictions have a higher median entropy. **Bottom-Left:** Reliability curve showing significant miscalibration, especially at low probabilities. **Bottom-Right:** Performance vs. retention when filtering predictions by an entropy threshold. Accuracy remains high as uncertain samples are removed, but retention drops quickly.